



M1.3 Unit Overview

Systems of Linear Equations and Inequalities

Students solve *systems of linear equations* and *inequalities*. They solve systems of linear equations using *elimination*, *substitution*, and graphing. They graph systems of linear inequalities and identify solutions. They write systems of linear equations and systems of inequalities to represent real situations. They interpret solutions to these systems in context. Students prove the slope criteria for parallel lines and *perpendicular* lines and derive the *distance formula*. They use coordinates, equations, and formulas to solve problems, including determining the perimeter and area of polygons.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Systems of Equations, Modeling with Functions, Comparing Models, CC4: Building with Triangles, Transformations and Congruence, NS: Financial Literacy

Explore: A Community Fundraiser

(Optional) Launch the unit with a non-routine task to investigate the question, "How do different constraints affect profit?"

Sub-Unit 1: Systems of Equations (Lessons 1–8 + Practice Day)

- Solve systems of linear equations using elimination, substitution, and graphing.
- Write systems of linear equations to represent constraints or meet given criteria.
- Language Goal:** Interpret solutions to systems of equations in context.

Sub-Unit 2: Systems of Inequalities (Lessons 9–12 + Practice Day + Quiz)

- Graph systems of linear inequalities and identify solutions.
- Write systems of linear inequalities to represent constraints or meet given criteria.
- Language Goal:** Interpret solutions to systems of inequalities in context.

Sub-Unit 3: Proving Geometric Theorems Algebraically (Lessons 13–17 + Practice Day)

- Prove and use the slope criteria for parallel and perpendicular lines to write equations.
- Derive and use the distance formula to solve problems.
- Language Goals:** Justify relationships between parallel and perpendicular lines.

Contextual Vocabulary

alligator, fox, flower, lizard, configurations, handholds, patchwork quilt, quilt squares, community meals, vegetarian, farm-to-table, floor plan, pick-up station, waiting area

Standards

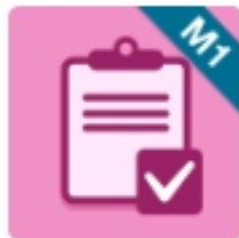
Addressing	A-CED.2	A-CED.3	A-REI.11	A-REI.12	A-REI.5	A-REI.6	A–SSE.1.a	F-IF.7	F-IF.7a	G-CO.1	G-CO.2	G-GPE.4	G-GPE.5
	G-GPE.7	N-Q.1	N-Q.2	SMP.1	SMP.2	SMP.3	SMP.4	SMP.5	SMP.6	SMP.7	SMP.8		

Vocabulary

- altitude
- conjecture
- distance formula
- elimination
- line segment
- opposite reciprocals
- perpendicular
- solution to a system of equations
- solution to a system of inequalities
- substitution
- system of equations
- system of inequalities

Review Vocabulary

- boundary line
- coefficient
- constraint
- distributive property
- equivalent equations
- equivalent systems
- half-plane
- infinitely many solutions
- no solution
- opposite(s)
- parallel lines
- parallelogram
- point of intersection
- profit
- Pythagorean theorem
- reciprocal
- slope (rate of change)
- slope-intercept form
- solution
- solution region
- standard form (of a linear equation)
- substitute
- sum
- translation
- x-coordinate
- x-intercept
- y-coordinate
- y-intercept



M1.3 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete using a scientific calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

📄 Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Linear Equations (Grade 8), Systems of Equations

Standards

Assessing 8.EE.7 8.EE.8 8.EE.8.b A-CED.3 A-REI.10 A-REI.12 A-REI.3 SMP.2 SMP.6 SMP.7



Explore: A Community Fundraiser



How do different constraints affect profit?

OPTIONAL This lesson is optional as the standards are addressed in other lessons within this course.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Ideas: CC2: Systems of Equations, NS: Financial Literacy

Review Vocabulary

- constraint
- profit

Today's Goals

- Determine how many participants are needed to meet certain fundraising goals within given constraints.
- Language Goal:** Explain potential modifications to fundraising earnings and costs to increase profit. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Materials

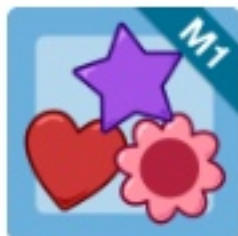
scientific calculators (optional)

Standards

Building On 8.EE.8.b 8.EE.8.c A-CED.3 Building Toward A-CED.3 A-REI.11 A-REI.6 SMP.2 SMP.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11



Shape It Up

Lesson 1: Introduction to Systems of Equations

Let's use reasoning to solve shape puzzles.

OPTIONAL This lesson is optional as it addresses standards from a prior course, supporting students in connecting to prior knowledge.

🗉 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to revise their thinking and Challenge Creator supports students in creating and sharing challenges with classmates.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Use reasoning to solve shape puzzles to prepare students for solving systems of equations in this unit.
- **Language Goal:** Explain a strategy for solving shape puzzles to prepare students for solving systems of equations in this unit. (*Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.EE.8 A-REI.1 Building Toward A-REI.6 SMP.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2

Review Vocabulary

- solution
- substitute
- sum

Instructional Routines

- Notice and Wonder
- Collect and Display (MLR2)



Eliminating Shapes

Lesson 2: Introduction to Elimination

Let's solve systems of equations by adding or subtracting the equations to eliminate a variable.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to revise their thinking. Teachers can use the dashboard's Teacher View to quickly access student thinking and identify students who may need additional supports or stretches.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve systems of equations by adding or subtracting the equations to eliminate a variable.
- Language Goal:** Identify when to use addition or subtraction to eliminate a variable. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 8.EE.8 A-REI.1 Addressing A-REI.6 A-SSE.1.a SMP.3 SMP.7 Building Toward A-CED.3 A-REI.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.8

Vocabulary

- elimination
- solution to a system of equations
- system of equations

Review Vocabulary

- coefficient
- opposites
- standard form

Instructional Routines

- Number Talk
- Compare and Connect (MLR7)
- Decide and Defend



Process of Elimination

Lesson 3: Elimination Using Equivalent Equations

Let's create equivalent equations to eliminate a variable.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve a system of equations using an elimination strategy.
- Language Goal:** Explain how replacing an equation in a system with one that is equivalent creates a new system that has the same solution. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

equivalent systems

Additional Materials

Activity 3 Cards (one per pair, pre-cut); coloring tools (as needed); Show What You Know Sheet

Standards

Building On 8.EE.8 A-REI.1 Addressing A-REI.5 A-REI.6 A-SSE.1.a SMP.3 SMP.6 SMP.7 Building Toward A-CED.3 A-REI.5

Language Standards

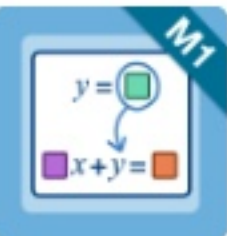
ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.6

Review Vocabulary

- distributive property
- equivalent equations
- opposite

Instructional Routines

- Compare and Connect (MLR7)



Solution by Substitution

Lesson 4: Solving Systems by Substitution

Let's use substitution to solve systems of equations.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Repeated Challenges offer Responsive Feedback to support students in developing fluency in solving systems of equations.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve systems of equations using substitution.
- Language Goal:** Describe a strategy for solving systems of equations using substitution. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Vocabulary

- substitution

Review Vocabulary

- distributive property

Instructional Routines

- Notice and Wonder
- Critique, Correct, Clarify (MLR3)
- Think-Pair-Share

Standards

Building On 8.EE.8 Addressing A-REI.6 SMP.1 SMP.5 SMP.6 SMP.7 Building Toward A-REI.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6



Lizard Lines

Lesson 5: Graphing Systems of Linear Equations

Let's explore systems of equations using graphs.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions, Responsive Feedback, and the Desmos Graphing Calculator to make connections across multiple representations and revise their thinking.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Solve a system of linear equations using a graph.
- Language Goal:** Analyze the structure of a system of equations to justify whether the system has zero, one, or infinitely many solutions.
(Writing, Speaking, and Listening)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Standards

Building On 8.EE.8.a 8.EE.8.a 8.EE.8.b A-REI.10 Addressing A-REI.10 A-REI.11 A-REI.6 SMP.2 SMP.2 SMP.3 SMP.3 SMP.7 SMP.7

Building Toward A-REI.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.8 ELD.PII.9-10.4

Review Vocabulary

- infinitely many solutions
- no solution
- parallel lines
- point of intersection
- slope (rate of change)
- slope-intercept form
- y-intercept

Instructional Routines

- Notice and Wonder
- Compare and Connect (MLR7)
- Decide and Defend



City Development

Lesson 6: Solving Graphically and Symbolically

Let's solve systems of equations using multiple methods and interpret solutions in context.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations, Modeling with Functions

Today's Goals

- Solve systems of equations graphically and symbolically.
- Language Goal:** Interpret the solution to a system of equations in context. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

CA EP&C Principle 5

Additional Materials

coloring tools (optional); scientific calculators (as needed); straightedges (as needed), Show What You Know Sheet

Standards

Building On 8.EE.8 A-REI.10 Addressing A-REI.6 F-IF.7 F-IF.7a N-Q.1 SMP.1 SMP.2 Building Toward A-CED.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.6 ELD.PI.9-10.8

Review Vocabulary

- constraint
- slope-intercept form
- standard form (of a linear equation)
- x-intercept
- y-intercept

Instructional Routines

- Think-Pair-Share
- Discussion Supports (MLR8)



Bus Systems

Lesson 7: Writing and Solving Systems of Equations

Let's model real-world situations using systems of equations.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Systems of Equations, Modeling with Functions, Comparing Models

Today's Goals

- Represent a situation with a system of equations.
- Solve systems of equations graphically and symbolically.
- **Language Goal:** Interpret the meaning of a system of equations and its solutions in context in order to make recommendations. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

coloring tools (as needed); Desmos Graphing Calculator (as needed); graph paper (as needed); straightedges

Standards

Building On 8.EE.8 A-CED.1 A-REI.3 Addressing A-CED.2 A-CED.3 A-REI.6 F-IF.7 F-IF.7a SMP.1 SMP.2 SMP.3 SMP.5

Building Toward A-REI.12 A-REI.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- constraints

Instructional Routines

- Three Reads (MLR6)
- Think-Pair-Share



Electric Line Zapper

Lesson 8: Strategically Solving Systems of Linear Equations

Let's solve systems of equations strategically.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback in order to make connections across multiple representations and revise their thinking.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Develop fluency in solving systems of linear equations using substitution and elimination.
- Language Goal:** Describe how to look for and make use of structures in equations to choose a solving strategy. (*Speaking and Listening*)
- Language Goal:** Explain the connection between a false statement obtained algebraically and no solution to the system. (*Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Standards

Building On [8.EE.8](#) Addressing [A-REI.6](#) [SMP.5](#) [SMP.7](#) Building Toward [A-REI.7](#)

Language Standards

[ELD.PI.9-10.1](#) [ELD.PI.9-10.12](#) [ELD.PI.9-10.3](#)

Instructional Routines

- Compare and Connect (MLR7)



M1.3 Practice Day 1



Students practice the concepts, skills, and strategies developed in Lessons 1–8. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Systems of Equations, Modeling with Functions

Additional Materials

Scavenger Hunt Sheets (one single-sided set); scientific calculators (optional)

Standards

Addressing A-CED.3 A-REI.6 F-IF.7 SMP.2 SMP.5 SMP.7



Quilts

Lesson 9: Introduction to Systems of Inequalities

Let's explore what solutions to a system of inequalities mean.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students create their own patch design that is replicated into a larger quilt using a digital interaction.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, NS: Financial Literacy

Today's Goals

- Determine if a point is a solution to a *system of inequalities* using a graph.
- Language Goal:** Describe how ordered pairs relate to constraints in a situation. (*Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 7.EE.4.b A-CED.1 Addressing A-CED.3 SMP.2 SMP.3 Building Toward A-REI.12

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- system of inequalities

Review Vocabulary

- boundary line

Instructional Routines

- Decide and Defend



Seeking Solutions

Lesson 10: Solutions to Systems of Inequalities

Let's explore strategies for determining the solution region for a system of inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? The digital interactions support students in visualizing the solution region for a system of inequalities.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Determine the solution region of a system of inequalities given the boundary lines.
- Language Goal:** Describe how to determine the solution region of a system of inequalities on a graph. (*Speaking, Listening, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

calculator (optional); coloring tools (as needed)

Standards

Building On A-REI.10 A-REI.12 Addressing A-REI.12 F-IF.7 SMP.3 SMP.5 SMP.6 Building Toward A-CED.3 A-REI.12

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.5

Vocabulary

- solution (to a system of inequalities)

Review Vocabulary

- boundary line
- no solution
- solution region

Instructional Routines

- Stronger and Clearer Each Time (MLR1)
- Collect and Display (MLR2)
- Critique, Correct, Clarify (MLR3)



Boundaries and Shading

Lesson 11: Graphing Systems of Inequalities

Let's write and graph systems of inequalities.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and an assignable Digital Companion needed.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Systems of Equations

Today's Goals

- Write a system of inequalities when given a graph.
- Graph the solution region to a system of inequalities.
- **Language Goal:** Describe a strategy for graphing the solution region to a system of inequalities. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

half-plane

Additional Materials

Activity 1 Cards (one set per pair, pre-cut); coloring tools; straightedges (optional)

Standards

Building On A-CED.1 A-REI.12 Addressing A-REI.12 F-IF.7 F-IF.7a SMP.6 SMP.7 Building Toward A-CED.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11

Review Vocabulary

- slope-intercept form
- solution region
- standard form (of a linear equation)
- x-intercept
- y-intercept

Instructional Routines

- Critique, Correct, Clarify (MLR3)



Community Meals

Lesson 12: Using Systems of Inequalities to Make Decisions

Let's use systems of inequalities to model real-world situations.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and an assignable Digital Companion needed.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Systems of Equations, NS: Financial Literacy

Review Vocabulary

- constraint

Today's Goals

- Write systems of inequalities to represent constraints in context.
- Language Goal:** Interpret the solutions to a system of inequalities in context in order to make decisions. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

coloring tools (optional)

Standards

Building On A-CED.1 A-REI.3 Addressing A-CED.3 A-REI.12 F-IF.7 N-Q.1 N-Q.2 SMP.1 SMP.4 Building Toward N-Q.1 N-Q.2 N-Q.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3



M1.3 Practice Day 2

Students practice the concepts, skills, and strategies developed in Lessons 1–12. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Systems of Equations, Modeling with Functions, NS: Financial Literacy

Additional Materials

Option 1: Solve and Swap

- Problem Cards (one copy per seven students, pre-cut)
- scientific calculators (optional)

Option 2: Partner Practice

- Problem Cards (one set per pair, pre-cut)
- scientific calculators (optional)

Standards

Addressing A-CED.3 A-REI.12 A-REI.6 F-IF.7 SMP.2 SMP.5 SMP.7

Language Standards

ELD.PI.9-10.12



M1.3 Sub-Unit Quiz



Learn about your students' understanding of the concepts and skills so far in this unit. This assessment is intended for students to complete using a scientific calculator.

Note: A paper option of this digital quiz is also available.

🗉 Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Systems of Equations, Modeling with Functions

Standards

Assessing

A-CED.3

A-REI.11

A-REI.12

A-REI.6

F-IF.7

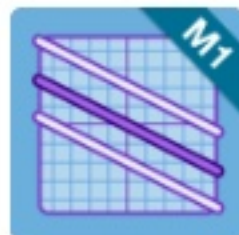
F-IF.7a

SMP.2

SMP.3

SMP.5

SMP.7



Parallel Lines on the Coordinate Grid

Lesson 13: Proving Slope Criteria for Parallel Lines

Let's use coordinates to investigate parallel lines.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to make sense of their slope calculations and visualize equations of lines, allowing them to revise their thinking.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, CC4: Building with Triangles, Transformations and Congruence

Today's Goals

- Prove algebraically the slope criteria for parallel lines.
- Use the parallel line slope criteria to write equations of lines.
- **Language Goal:** Explain how translations and the slope formula justify that parallel lines have the same slope. (*Speaking, Listening, and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Standards

Building On 8.G.1 A-SSE.1.a Addressing G-CO.1 G-CO.2 G-GPE.4 G-GPE.5 N-Q.1 SMP.1 SMP.6 SMP.7 Building Toward G-GPE.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12

Review Vocabulary

- parallelogram
- translation

Instructional Routines

- Think-Pair-Share
- Notice and Wonder
- Collect and Display (MLR2)



Perpendicular Lines on the Coordinate Grid

Lesson 14: Proving Slope Criteria for Perpendicular Lines

Let's investigate the slopes of perpendicular lines.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, CC4: Building with Triangles, Transformations and Congruence

Today's Goals

- Prove that the slopes of *perpendicular* lines are *opposite reciprocals*.
- Use slopes of *perpendicular lines* to solve problems.
- **Language Goal:** Justify whether two lines are *perpendicular*. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Show What You Know Sheet

Standards

Building On 8.G.1 8.G.3 Addressing G-CO.1 G-CO.2 G-GPE.4 G-GPE.5 N-Q.1 SMP.1 SMP.3 SMP.7 Building Toward G-GPE.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3

Vocabulary

- conjecture
- opposite reciprocal
- perpendicular

Review Vocabulary

- opposite
- reciprocal

Instructional Routines

- Notice and Wonder
- Decide and Defend
- Collect and Display (MLR2)



It's All on the Line

Lesson 15: Determining Equations of Lines

Let's work with both parallel and perpendicular lines.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, CC4: Transformations and Congruence

Today's Goals

- Write an equation for a line that travels through a given point and is either parallel or perpendicular to a given line.
- Language Goal:** Explain why two lines that are perpendicular to the same line must be parallel to each other. (*Speaking and Listening, Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

straightedges, Show What You Know Sheet

Standards

Building On 8.EE.6 8.EE.8 A-REI.10 Addressing A-CED.2 G-GPE.5 SMP.6 SMP.7 Building Toward A-REI.6

Language Standards

ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- parallel
- perpendicular
- reciprocal
- translation



Going the Distance

Lesson 16: Using Coordinates to Find Length

Let's explore another way to find the length of the segment that connects two points.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC4: Building with Triangles, Transformations and Congruence

Today's Goals

- Derive the distance formula from the Pythagorean theorem.
- Use the distance formula to find the length of the segment that connects two points.
- Language Goal:** Compare the Pythagorean theorem to the *distance formula*. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Show What You Know Sheet, scientific calculators (as needed)

Standards

Building On 8.G.8 Addressing G-CO.1 G-GPE.4 SMP.3 SMP.6 SMP.8 Building Toward G-GPE.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- distance formula
- line segment

Review Vocabulary

- coordinates
- Pythagorean theorem
- x-coordinate
- y-coordinate

Instructional Routines

- Critique, Correct, Clarify (MLR3)
- Notice and Wonder



Restaurant Math

Lesson 17: Determining Perimeter and Area on the Coordinate Grid

Let's explore the perimeter and area using coordinates.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Idea: CC4: Building with Triangles

Today's Goals

- Use coordinates and the distance formula to compute perimeters and areas of polygons.
- **Language Goal:** Describe the information needed and the steps for determining the perimeter and area of shapes on a coordinate grid.
(Reading, Writing, Speaking, and Listening)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

capacity, decomposed, superimposed

Additional Materials

scientific calculators

Standards

Building On 8.G.8 Addressing G-GPE.7 SMP.1 SMP.4 SMP.6 Building Toward G-GPE.4

Language Standards

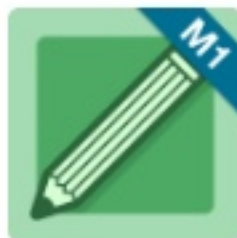
ELD.PI.9-10.1 ELD.PI.9-10.3 ELD.PI.9-10.6 ELD.PI.9-10.8

Vocabulary

- altitude

Instructional Routines

- Notice and Wonder
- Think-Pair-Share



M1.3 Practice Day 3



Students practice the concepts, skills, and strategies developed in Lessons 1–17. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Systems of Equations, Modeling with Functions, CC4: Building with Triangles, Transformations and Congruence, NS: Financial Literacy

Materials

Option 1: Solve and Swap

- Problem Cards (one per seven students, pre-cut)
- scientific calculators (optional)
- coloring tools (optional)

Option 2: Partner Practice

- Problem Cards (one set per pair, pre-cut)
- scientific calculators (optional)
- coloring tools (optional)

Standards

Addressing A-CED.3 A-REI.12 A-REI.6 G-GPE.5 G-GPE.7 SMP.2 SMP.6 SMP.7



M1.3 Performance Task



Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Modeling with Functions, Systems of Equations, CC4: Transformations and Congruence, Shapes in Structures

Standards

Assessing

A-CED.3

G-GPE.4

G-GPE.5

G-GPE.7

SMP.3

SMP.4



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

🗉 Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.